

SARIWON, Korea, Democratic People's Republic ofWMO: **470650**Lat: **38.517N**Lon: **125.767E**Elev: **52**StdP: **100.70**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.9	-20.6	0.6	-11.0	-18.3	0.8	-9.5	12.0	-6.6	10.7	-6.6	2.3	320	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	31.6	25.8	30.4	24.8	29.3	24.1	27.0	30.5	26.2	29.3	25.5	28.2	2.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.0	21.5	29.2	25.2	20.5	28.3	24.6	19.7	27.6	85.4	30.6	81.9	29.7	78.7	28.2	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.8	7.5	-15.9	33.7	2.9	1.6	-17.9	34.9	-19.6	35.8	-21.2	36.8	-23.3	37.9		
			-16.4	28.3	2.6	0.9	-18.3	28.9	-19.8	29.5	-21.3	30.0	-23.2	30.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-4.7	-1.9	4.3	11.4	17.6	22.3	25.1	25.3	20.5	13.8	5.5	-2.0
(7)		DBStd	11.05	4.12	3.98	3.69	3.72	2.70	2.16	2.06	2.30	2.54	3.54	4.59	4.24
(8)		HDD10.0	1526	454	333	182	27	0	0	0	0	10	147	372	
(9)		HDD18.3	3141	713	567	436	208	46	2	0	0	10	144	386	630
(10)		CDD10.0	2073	0	0	4	71	236	368	467	475	315	128	11	0
(11)		CDD18.3	648	0	0	0	1	24	119	208	217	75	3	0	0
(12)		CDH23.3	4597	0	0	0	9	182	706	1549	1769	364	16	0	0
(13)		CDH26.7	1174	0	0	0	0	24	163	414	530	42	1	0	0
(14)	Wind	WSAvg	2.4	2.4	2.7	3.0	2.8	2.8	2.2	2.1	1.9	1.7	2.0	2.4	2.4
(15)	Precipitation	PrecAvg	907	10	14	19	48	79	92	267	209	97	44	36	17
(16)		PrecMax	1788	27	48	52	116	219	272	595	568	377	120	109	45
(17)		PrecMin	400	1	0	0	1	7	23	75	23	12	2	5	2
(18)		PrecStd	279	7	11	14	27	52	59	131	129	92	34	25	11
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	6.7	11.3	17.7	24.7	29.1	31.2	33.4	33.6	29.3	25.2	18.2	10.0	
(20)		0.4% MCWB	4.2	7.1	11.1	15.6	18.6	22.4	26.4	27.0	22.7	18.3	14.1	7.1	
(21)		2% DB	4.5	7.9	14.8	22.2	26.8	29.8	31.5	31.9	27.9	22.9	15.9	7.7	
(22)		2% MCWB	2.2	4.3	9.1	14.4	18.0	21.7	25.7	26.7	21.8	16.9	12.2	5.5	
(23)		5% DB	2.9	5.9	12.6	20.2	25.3	28.6	30.2	30.8	26.8	21.2	14.1	5.7	
(24)		5% MCWB	0.9	2.9	7.9	13.0	17.2	21.2	25.1	25.9	21.0	16.1	11.1	3.7	
(25)		10% DB	1.3	4.2	10.6	18.2	23.7	27.2	29.0	29.5	25.6	19.6	12.3	4.0	
(26)		10% MCWB	-0.4	1.7	6.4	12.2	16.5	20.6	24.7	25.1	20.3	15.1	9.6	2.2	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	4.8	7.7	11.9	18.1	21.0	24.3	27.8	28.3	24.5	19.6	15.3	8.1	
(28)		0.4% MCDB	6.2	9.9	16.1	22.9	25.7	29.1	31.1	31.9	27.4	23.6	17.1	9.1	
(29)		2% WB	2.6	4.7	10.1	15.5	19.4	23.2	26.7	27.3	23.4	18.1	13.1	5.7	
(30)		2% MCDB	4.0	7.2	13.8	20.2	24.5	27.5	30.1	30.8	26.1	21.1	15.2	7.4	
(31)		5% WB	1.1	3.2	8.5	14.1	18.4	22.4	26.0	26.6	22.4	17.0	11.4	3.8	
(32)		5% MCDB	2.6	5.5	11.9	18.7	23.3	26.5	29.1	29.9	25.3	20.2	13.8	5.5	
(33)		10% WB	-0.3	1.9	6.8	12.8	17.5	21.7	25.3	25.9	21.4	15.8	9.8	2.3	
(34)		10% MCDB	1.2	3.9	9.9	17.1	22.1	25.6	28.0	28.8	24.5	18.9	11.9	3.9	
(35)	Mean Daily Temperature Range	MDBR	7.7	8.2	9.0	10.4	10.3	8.7	6.4	7.1	9.2	9.8	7.9	7.0	
(36)		5% DB MCDBR	9.1	10.5	12.2	13.2	13.2	11.2	8.6	8.4	10.2	11.3	10.0	8.6	
(37)		5% DB MCWBR	7.3	7.7	8.0	7.1	6.4	5.2	4.3	4.3	5.2	6.1	6.9	7.0	
(38)		5% WB MCDBR	8.3	9.4	11.1	11.1	10.9	9.0	6.9	7.3	8.3	9.8	8.9	8.1	
(39)		5% WB MCWBR	7.1	7.5	8.0	6.9	6.2	4.9	4.1	4.2	5.1	6.0	7.0	6.9	
(40)	Clear-Sky Solar Irradiance	taub	0.420	0.487	0.550	0.613	0.626	0.707	0.680	0.580	0.511	0.523	0.495	0.431	
(41)		taud	1.929	1.814	1.718	1.638	1.658	1.564	1.669	1.882	1.980	1.843	1.850	1.934	
(42)		Ebn at Noon	700	703	705	692	695	640	653	711	735	661	613	651	
(43)		Edh at Noon	138	178	216	248	248	272	243	192	164	170	150	129	
(44)	All-Sky Solar Radiation	RadAvg	2.44	3.20	4.33	5.05	5.63	5.20	4.10	4.41	4.31	3.53	2.36	2.08	
(45)		RadStd	0.20	0.24	0.35	0.36	0.53	0.52	0.55	0.54	0.39	0.22	0.24	0.15	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47)	Regional (28 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +96
										(m) +55

Nomenclature: See separate page